

Hanul Jeon

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Academic Positions

Postdoctoral Researcher

Vienna University of Technology, Project Leader: Juan P. Aguilera

Sep 2026 –

Education

Ph.D.

Cornell University, Advisor: Justin Moore and James Walsh

Dissertation: *Proof theory for higher pointclasses*

Mathematics

Aug 2021 – May 2026

Visiting Scholar

Vienna University of Technology, Supervisor: Juan P. Aguilera

Jan 2025 – Jul 2025

M.Sc.

Seoul National University, Advisor: Otto van Koert

Thesis: *Constructive Ackermann's interpretation*

Mathematics

Mar 2017 – Feb 2021

B.Sc.

Sunkyunkwan University

Mathematics

Mar 2013 – Feb 2017

Experience

Republic of Korea Air Force

Compulsory military service

Aug 2017 – Jul 2019

Publications

- [1] Kinkar Ch Das, Han-ul Jeon, and Nenad Trinajstić. “Comparison between the Wiener index and the Zagreb indices and the eccentric connectivity index for trees”. In: *Discrete Applied Mathematics* 171 (2014), pp. 35–41.
- [2] Hanul Jeon. “Constructive Ackermann's interpretation”. In: *Annals of Pure and Applied Logic* 173.5 (2022), p. 103086.
- [3] Hanul Jeon and Richard Matthews. “Very large set axioms over constructive set theories”. In: *The Bulletin of Symbolic Logic* 30.4 (2024), pp. 455–535.
- [4] Hanul Jeon. “On Separating Wholeness Axioms”. In: *Journal of Symbolic Logic* 91.1 (2026), pp. 369–392.
- [5] Hanul Jeon. “The proof-theoretic strength of Constructive Second-order set theories”. In: *Notre Dame Journal of Formal Logic* 66.3 (2025), pp. 413–431.
- [6] Hanul Jeon and James Walsh. *Generalized ordinal analysis and reflection principles in set theory*. 2023. arXiv: 2312.12859 [math.LO].
- [7] Hanul Jeon. *The behavior of higher proof theory I: Case Σ_2^1* . 2024. arXiv: 2406.03801 [math.LO].
- [8] Hanul Jeon. “On a cofinal Reinhardt embedding without Powerset”. In: *Fundamenta Mathematicae* 270.2 (2025), pp. 177–200.
- [9] Hanul Jeon. *Proof-theoretic dilator and intermediate pointclasses*. 2025. arXiv: 2501.11220 [math.LO].
- [10] Hanul Jeon. *Martin's measurable dilator*. 2025. arXiv: 2503.12713 [math.LO].
- [11] Hanul Jeon et al. *Ranking theories via encoded β -models*. 2025. arXiv: 2503.20470 [math.LO].
- [12] Hanul Jeon. *Dilator-based analysis of KP*. 2026. arXiv: 2608.29157 [math.LO].

Teaching Experiences

Teaching assistant

Introduction to Analysis

Spring 2026

Grader

Mathematical Logic

Fall 2024

Grader

History of Mathematics

Spring 2023

Grader

Applied logic

Fall 2023

Grader

Introduction to Analysis

Spring 2023

Grader

Mathematical Logic

Fall 2022

Tutor

Sublime Learning Community Tutoring

Spring, Summer, Fall 2020

Covered Mathematics and its practice (Spring 2020) and Mathematics for biological scientists (Summer 2020, Fall 2020).

Grader

Mathematics for ecologists

Fall 2020

Teaching Assistant and Grader

Advanced Mathematics and its practice II.

Fall 2020

Equivalent to Multivariable Calculus Honors. Managed students' presentation.

Teaching Assistant and Grader

Logic and Set theory

Spring 2020

Teaching Assistant and Grader

Mathematics and its practice II.

Fall 2019, Summer 2020, Winter 2020

Equivalent to Multivariable Calculus. Managed an exercise course.

Teaching Assistant and Grader

Mathematics and its practice I.

Spring 2017, Spring 2019

Equivalent to Calculus. Managed an exercise course.

Tutor

Sunkyun Tutoring for Advanced Algebra

Spring 2016

Covered Chapters 4 and 5 of Hungerford's *Algebra*

Grants and Fellowships

Hutchinson Fellowship 2024–2025:

Fall 2025

OeAD Ernst Mach Grant: 6,250 EUR

Feb 2025 – Jul 2025

Cornell Research Travel Grant: 1100 USD (for Austrian visit)

Oct 2024

The National Scholarship for Science and Engineering:

Mar 2013 - Dec 2016

Talks

On the hierarchy of theories

KIAS Logic Seminar

Korea Institute For Advanced Study

January 2026

On proof-theoretic dilator and Pohlers' characteristic ordinals

Universitäts Wien

Reverse Mathematics and Higher Computability Theory Workshop

July 2025

Dilators in Descriptive set theory

TU Wien

TU Wien Computational Logic Seminar

April 2025

On proof-theoretic dilator and Pohlers' characteristic ordinals

Changwon University

4th Korea Logic day

January 2025

On a cofinal Reinhardt embedding without Powerset

CUNY Graduate Center

CUNY Set theory seminar

October 2024

On Separating Wholeness Axioms

2024 North American ASL Annual Meeting

Proof theory for higher pointclasses

UPenn Logic and Computation Seminar

On Separating Wholeness Axioms

3rd Korean Logic Day

On Separating Wholeness Axioms

Cornell Logic Seminar

Very large set axioms over constructive set theories

Cornell Logic Seminar

Consequences of the axiom of choice

Olivetti Club

Goodstein's theorem

Madmathematics Seminar

What is forcing?

Madmathematics Seminar

A Short introduction to mathematical logic

Madmathematics Seminar

Iowa State University

May 2024

University of Pennsylvania

April 2024

Virtual

Jan 2024

Cornell University

May 2023

Cornell University

Apr 2022

Cornell University

Mar 2022

Seoul National University

Oct 2016

Seoul National University

Nov 2015

Sogang University

Jan 2014